Radial light profiles for galaxy classification, and DEBBIE: 161,721 galaxy candidates with no prior identification in DESI Legacy Imaging Surveys DR10

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Abstract

Galaxy catalogues are usually checked by eye, object by object, which is impractical at 10^5 entries. We measure purity on a sample instead. Each source is described by its light profile in 24 concentric annuli and 32 position angles in g, r and z, and classified by gradient-boosted trees. On the DeepShadows benchmark this reaches AUC 0.9810 (95% CI 0.9765–0.9852), against 0.9098 for six conventional catalogue features with the same classifier and 0.9740 for the published convolutional network.

Applied to the DESI Legacy Imaging Surveys DR10, the method gives DEBBIE, 161,721 galaxy candidates over 12,900 deg^2, selected by cuts on the survey's catalogue. None has a prior identification as a galaxy in 1.4×10^8 catalogued objects, NED, SIMBAD, SMUDGes or DESI DR1. Inspecting a stratified sample finds a galaxy at the catalogued position for 89.1% of entries (86.4–91.6%). The trees and a recurrent network on the same profiles flag an estimated 7.2% of entries as doubtful, and purity among the rest is 90.7%, close to the SMUDGes pipeline before its visual check and well above the DES search at the same stage. DEBBIE is 7 times the size of the DES catalogue and 23 times that of SMUDGes. Measured on known galaxies, the cuts retain 56–89% of those with $r_{\rm eff} = 3$–$15''$ and fail above $15''$, and a filter against fragments of catalogued galaxies loses 29% of uncatalogued candidates. All redshifts are photometric, with median $z = 0.29$, so most entries are ordinary galaxies dimmed by distance. The catalogue, code and source table are released.

1 Introduction

Imaging surveys publish a catalogue of every source they detect, and those catalogues hold galaxies that no galaxy catalogue lists. Recovering them need not start from the pixels. Tanoglidis, Drlica-Wagner, et al. (2021) selected low-surface-brightness candidates from the Dark Energy Survey source catalogue, classified them by machine and inspected the survivors by eye, while SMUDGes (Zaritsky et al. 2019) runs its own detection on the imaging. Both checked every object they kept by eye, as did Thuruthipilly et al. (2024). At 10^5 entries one inspector cannot, so purity has to be measured on a sample and doubtful entries flagged. We start from the source catalogue of the DESI Legacy Imaging Surveys (Dey et al. 2019), measured with Tractor (Lang et al. 2016). Cuts on those measurements and a screen against existing identifications select the catalogue, an inspected sample measures its purity, and two classifiers flag doubtful entries.

Low-surface-brightness (LSB) galaxies are under-represented in every major galaxy catalogue, because of the selection functions of the surveys those catalogues come from (Zaritsky et al. 2019), and our selection window lies at low surface brightness for that reason. Tanoglidis, Ciprijanović, et al. (2021) trained convolutional networks ("DeepShadows") to separate LSB galaxies from imaging artefacts in the Dark Energy Survey, and their visually labelled sample is now a benchmark.

Convolutional networks are designed for problems whose useful features have to be learned. For a galaxy the useful features are known in advance, because its light is close to elliptically symmetric, and its departures from that symmetry, such as bars, arms and lopsidedness, are features in their own right. Binning the

light in concentric annuli puts the symmetry into the representation directly, so no network has to learn it. We test whether that representation, with a standard classifier, matches the pixel-based deep model. We compare several classifiers as a candidate set, in the manner of multi-model inference in ecology (Burnham and Anderson 2002), reporting every member and testing whether a combination predicts better than the best one.

The catalogue is DEBBIE (DR10 Ensemble of Boosted trees and Bidirectional recurrent networks for Identifying Extragalactic sources), named for the two classifiers that flag its doubtful entries (Section 5.6.2). No classifier decides which candidates enter: the cuts and the screening for prior identification define the catalogue, and the two classifiers flag likely contaminants without removing any. It holds 161,721 galaxy candidates in DESI Legacy Imaging Surveys DR10 with no prior identification as galaxies in the catalogues and services we queried. We measure its selection function on real catalogued galaxies in place of synthetic injections (Section 5.5), and the whole selection can be recomputed from a released 3.2 GB source extract.

Fig. 1 shows the steps and the count that leaves each one. Section 2 describes the survey and benchmark data and Section 3 the radial representation. Section 4 tests the representation on the benchmark, Section 5 presents the catalogue with its selection function and purity, Section 6 compares it with the DES and SMUDGes catalogues, and Section 7 lists the results.

2 Data

2.1 Survey data

We use the DR10 south *sweep* catalogues, which hold the survey's Tractor photometry: positions, morphological types, shape parameters, and forced photometry in g, r, z and the WISE bands. Every candidate is a source Tractor detected and measured, so the catalogue inherits Tractor's completeness, which Section 5.5 measures.

The main catalogue covers DR10 south at Galactic latitude $|b| \geq 30^\circ$. We processed all 874 sweep files in that region, an occupied area of about 12,900 deg^2. We stop at $|b| = 30^\circ$ because at lower latitude Galactic cirrus looks like diffuse galaxies and foreground extinction corrupts the surface-brightness selection. The sweeps at $12^\circ < |b| < 30^\circ$ yield 78,713 further candidates, released as a separately flagged extension that carries a cirrus-contamination caveat and enters none of the statistics reported here.

2.2 Benchmark data

For classifier comparisons we use the DeepShadows sample (Tanoglidis, Ciprijanović, et al. 2021), restricted to their objects and to their visually confirmed hard-negative class (`artifact_hard_visually_rejected`), the class their published AUC is measured on. Including the easier negatives, which an earlier support-vector-machine stage of their pipeline rejected, raises the AUC of every method by about 0.014, so we leave them out and the comparison with the published AUC stays like for like.

The published model was trained on 30,000 objects and ours on the 10,121 available to us. It also works on different pixels, because the CNN works on DES imagery with DES photometry while we measure the same sky positions in DR10 imaging with our own measurements. The two pipelines therefore share only their labels, and any gap between them mixes the classifier with the imaging and the training size. Section 4.2 makes the controlled comparison, with everything but the features held fixed.

3 Radial representation

We measure each source in 24 concentric annuli with radii spaced geometrically from 1 to 84″, each divided into 32 position-angle cells, in each of g, r and z (Fig. 2). For each annulus we keep the mean and the median over position angle and the amplitudes of the first four azimuthal Fourier modes, from a discrete Fourier

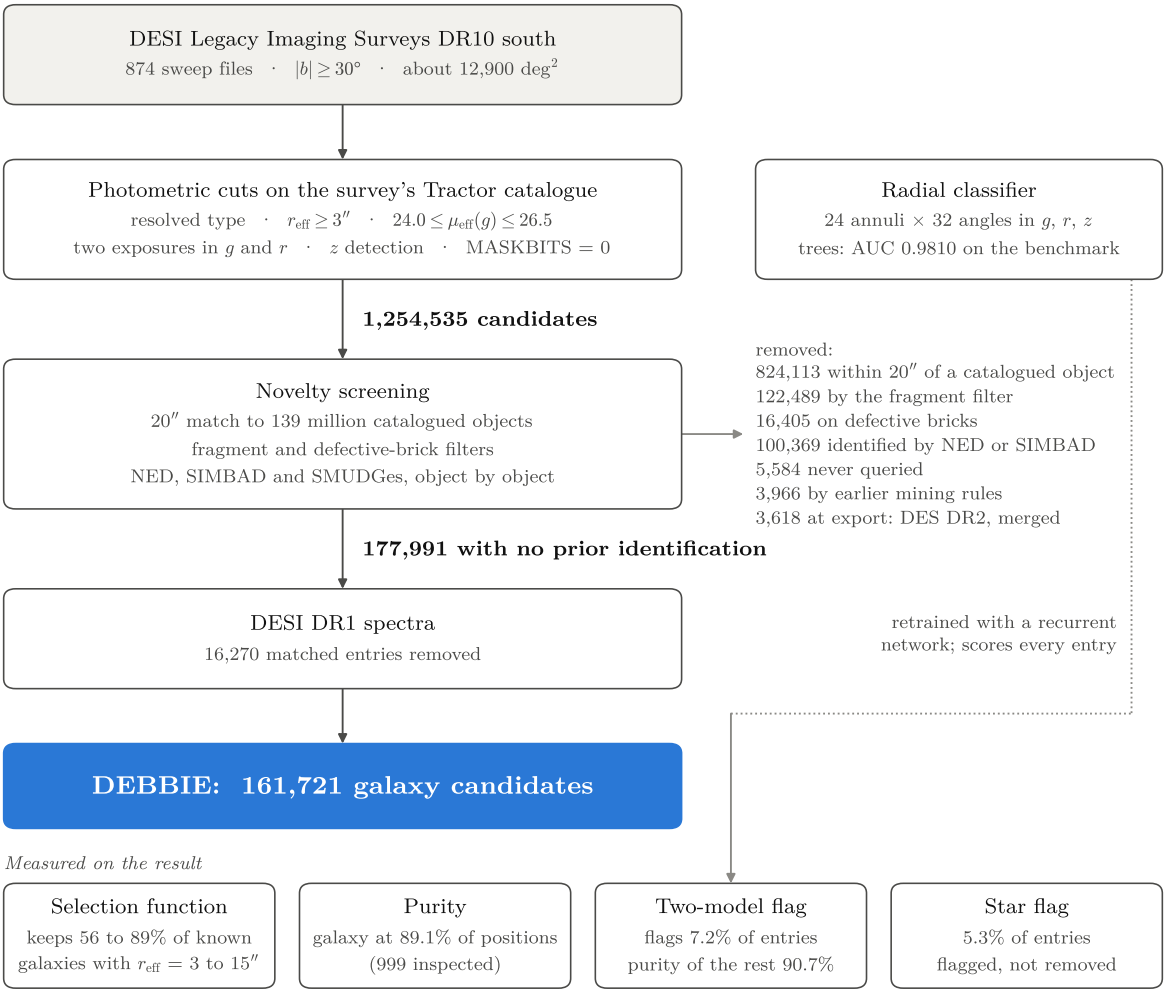


Figure 1: From survey to catalogue, with the number of candidates leaving each step. Photometric cuts and a screen against existing identifications define DEBBIE. The radial classifier of Section 4 enters the catalogue only through the two-model flag, which marks entries and removes none. The bottom row is measured on the result.

transform over the cells. Low-order Fourier terms have long been used in isophote analysis (Jedrzejewski 1987; Rix and Zaritsky 1995), and they encode lopsidedness and ellipticity without a shape fit. We keep the median as well as the mean because the median is robust to one bright neighbour in a single wedge.

We track the coverage of each cell against the count it would have if the frame were complete, so clipping at a brick edge registers as reduced coverage and does not look like a faint annulus. Each profile is extracted twice, at the catalogue position and at a flux-weighted refined centroid. We keep both, and the offset between the two centres is itself a feature, because a large offset flags blends and fragments. The feature definitions are in the released code.

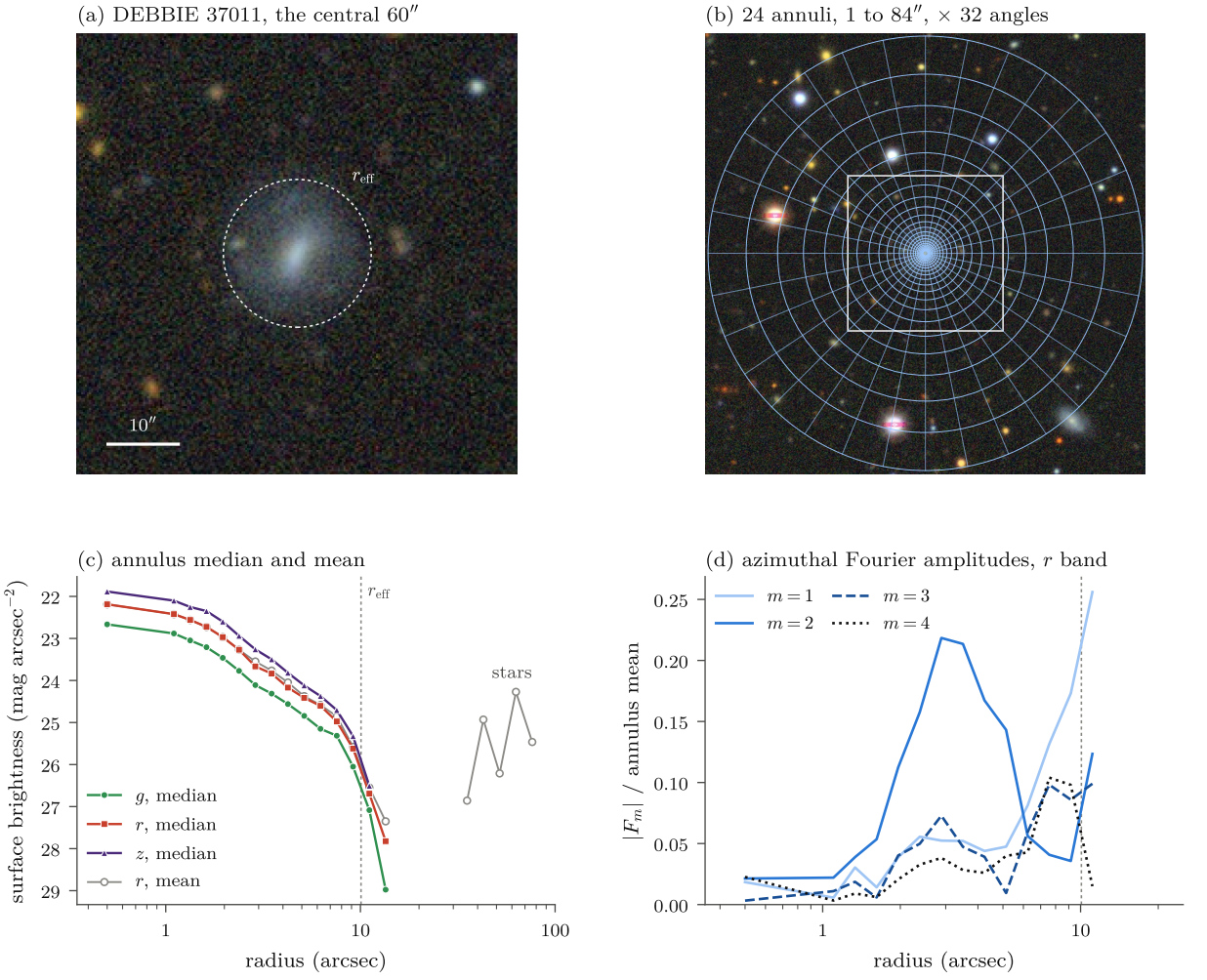


Figure 2: The representation of one catalogue galaxy, as the classifiers receive it. (a) DEBBIE 37011, accepted as a galaxy in the inspection of Section 5.6, with a circle at its effective radius. (b) The grid on the full 180″ field: 24 annuli with radii spaced geometrically from 1 to 84″, each cut into 32 position-angle cells; the square marks panel (a). (c) The median surface brightness of each annulus in g, r and z, out to where it falls below 3σ, and the r-band mean. Beyond 30″ stars in the outer annuli raise the mean but not the median. (d) Amplitudes of the first four azimuthal Fourier modes in r, relative to the annulus mean, out to where the median falls below 10σ; the $m = 2$ peak near 3″ is the elongated centre visible in (a). Panels (c) and (d) are computed from the calibrated images at 0.524″ per pixel. Colour images: Legacy Surveys / D. Lang (Perimeter Institute).

4 Classification results

4.1 Benchmark

On a balanced test set of 3,340 objects (1,670 per class), split by object and evaluated once, gradient-boosted trees (Chen and Guestrin 2016) on the radial representation reach AUC 0.9810, with a 2,000-resample bootstrap interval of [0.9765, 0.9852]. The published CNN reports 0.9740 (published range 0.9700–0.9740) on the same object and negative classes. The same classifier on six conventional catalogue features (profile monotonicity, concentration, signal-to-noise ratio, angular radius, central surface brightness and local imaging depth) reaches 0.9098 [0.9002, 0.9198] (Fig. 3). The classifiers are XGBoost 3.4.0 and scikit-learn 1.9.0. The trees use one setting throughout (400 trees of depth 4, learning rate 0.06, row and column subsampling 0.85), fixed before the benchmark was run, and the released code fixes the splits and seeds.

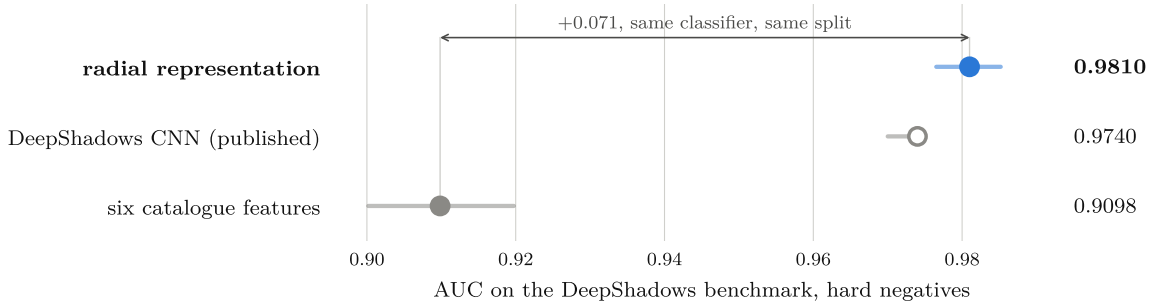


Figure 3: On the DeepShadows benchmark, the radial representation raises the AUC of one gradient-boosted classifier from 0.9098 to 0.9810, above the published CNN. Filled points: the same classifier on two inputs, on the same test split, with 2,000-resample bootstrap intervals. Open point: the published CNN with its published range; it was trained on more objects and on DES images (Section 2).

4.2 Representation versus architecture

We separate the representation from the model with a controlled ablation: one gradient-boosted classifier and four feature sets, so that only the classifier's input varies. The first test split is held out by sky position, so each field falls on one side of it; we call it the frozen spatial fold. On that fold, six conventional catalogue features score 0.9022, the per-annulus profile alone (144 features) 0.9645, that profile plus the six catalogue features (150) 0.9674, and the full flattened representation (1,023) 0.9775.

On five other test splits the same six-feature model scored between 0.861 and 0.911, so the choice of split alone moves an absolute AUC by about 0.05, nearly as much as the improvement from the six catalogue features to the full representation. We therefore repeated all four feature sets over five stratified folds of the benchmark sample (6,753 galaxies and 6,630 hard negatives) and scored each on every fold (Table 1). The ordering is the same on all five. The paired improvement from six catalogue features to the full representation is $+0.067 \pm 0.008$ and is at least $+0.060$ on every fold. Pairing matters because the difficulty of a split is common to all four and cancels in the difference, so the uncertainty that applies is the 0.008 spread of the paired improvement.

Table 1: The candidate models over five stratified folds of the benchmark sample, with identical folds throughout. In the first four rows only the input to the trees changes, and the representation raises the AUC by +0.067; the recurrent network on the annuli alone matches the trees, and the mean of the two is the score that flags catalogue entries.

model	features	mean AUC	sd
trees, six catalogue features	6	0.9116	0.0063
trees, per-annulus profile	144	0.9717	0.0029
trees, profile + six catalogue features	150	0.9731	0.0032
trees, full flattened representation	1,023	0.9790	0.0034
recurrent network on the annuli	sequence	0.9782	0.0032
mean of the last two		0.9808	0.0030

These five folds are randomly stratified, with fields shared between training and test, so their absolute values are optimistic next to the frozen spatial fold, as the two sets of numbers show. We therefore claim only the paired comparisons. The ablation shows that the representation carries information the six catalogue features lack.

The same folds compare architectures and test whether combining the candidates predicts better than the best one alone. A bidirectional recurrent network that takes the annuli in order of radius, without the six catalogue features, scores 0.9782. That is level with the trees on the full representation, with a paired difference of -0.0008 ± 0.0024 that favours the network on 2 folds of five.

Ecologists weight each member of a candidate set by its support and predict from the weighted set (Burnham and Anderson 2002). Astronomy uses information criteria and model averaging to choose between cosmological models (Liddle 2007; Parkinson and Liddle 2013), and classifier ensembles are common in source classification (Kim et al. 2015), including on these labels (Thuruthipilly et al. 2024). Trees have no parameter count, so in place of the Akaike information criterion we weighted each model by its deviance on held-out parts of the training fold. With about 10,700 training objects per fold one model takes more than 0.99 of the weight on every fold, so weighting reduces to selection. An unweighted mean of the trees and the recurrent network scores 0.9808, which is $+0.0017 \pm 0.0014$ above the trees and higher on all five folds, with a log-loss 7% lower. A mean over all five candidates is 0.0015 below the trees, because the weaker members lower it. Combining improves the AUC about 40 times less than the representation does.

5 Catalogue

5.1 Selection

From the sweep catalogues we select sources that meet all of the following: a resolved morphological type (`EXP`, `DEV`, `SER` or `REX`; Tractor assigns `SHAPE_R` = 0 to PSF sources, so this excludes point sources exactly); $r_{\rm eff} \geq 3''$; mean surface brightness within the effective radius, $\mu_{\rm eff}$, between 24.0 and 26.5 mag arcsec^{-2} in g; at least two exposures in both g and r; a z-band detection; and `MASKBITS` = 0. The z-band requirement is the weaker, single-epoch one because z-band epoch counts are lower over parts of the southern footprint. The faint edge of the surface-brightness window matters most. Above $\mu_{\rm eff} = 26.5$ Tractor increasingly fits large radii to noise, and in 48-object contact sheets of an early pool, 62% of which lay above that threshold, such sources mostly looked like background fluctuations.

5.2 Novelty screening

We reject any candidate within $20''$ of one of the 138,675,216 catalogued objects in a local index: the full HyperLEDA catalogue (Makarov et al. 2014), SDSS DR12 photometric galaxies to $r = 22.5$ between right

ascensions 120° and 235° (Alam et al. 2015), GLADE+ (Dálya et al. 2022), the 2MASS Extended Source Catalog (Jarrett et al. 2000), 2dFGRS (Colless et al. 2001), 6dFGS (Jones et al. 2009), SMUDGes (Zaritsky et al. 2019), and 28.5 million DES DR2 sources over part of the southern footprint (Abbott et al. 2021). The local screen runs at 4.5×10^5 positions s^{-1}, fast enough to screen every candidate. We then query each survivor against NED within 30″, counting any object of a galaxy type as an identification, against SMUDGes within 15″, and against SIMBAD within 15″, where any object at all counts against the candidate. A query that fails is repeated, and a failed check is never recorded as novelty. The queries ran in order of decreasing $r_{\rm eff}$ and did not reach 5,584 candidates, 5,548 of them below 3.05″, just above the size limit. These are excluded, so the screen can lose a real candidate but cannot create a spurious one.

Tractor splits extended sources into several fragments, so an outer fragment can sit more than 20″ from the catalogued position of its galaxy and pass that screen. We remove such fragments by rejecting any candidate with an object of the local index between 20″ and 45″. The filter also discards genuine candidates. At 20,000 random footprint positions, 22.2% have an index entry in that annulus, and among candidates that no catalogue lists within 20″ the filter removes 29% (Section 5.5.1). The selection function of Section 5.5 measures the photometric cuts alone and so leaves that loss out. When Tractor splits one galaxy into several rows that all survive, we merge them with a matching radius of $\max(6'', r_{\rm eff})$ and keep the largest member; 2,013 entries were merged.

A further filter removes spurious detections that cluster on defective imaging bricks, by rejecting any candidate with three or more other candidates within 1.5′. We exclude one region of 6′ radius at $(\alpha, \delta) = (144.79°, -2.79°)$ permanently, having confirmed it by eye.

5.3 What counts as an identification

No object in DEBBIE has a prior identification as an extragalactic object: none passes within the radii above of a catalogued galaxy, of a NED object typed G, GPair, GTrpl, GGroup, GClstr, QSO or AGN, or of any SIMBAD (Wenger et al. 2000) object. A designation from an automated survey is not an identification. Nearly all entries (99.3%) carry one, most often from WISE (Wright et al. 2010), so they were detected before but never typed, and we record every designation NED lists within 0.5′ of each position.

DESI DR1 (DESI Collaboration 2025) is a public spectroscopic release that covers this imaging footprint and selects its targets from the same parent photometry. Matching the candidate list against it at 2″ finds a reliable spectrum (ZWARN = 0) for 16,270 main-catalogue entries, 9.14% of the list, of which 16,014 are typed GALAXY, 116 QSO and 140 STAR. The galaxies and quasars have a published spectroscopic identification as extragalactic, which our definition counts as prior identification, and the stars are not galaxies. We remove all 16,270, which leaves 161,721 entries, and every number in this paper is computed on that set.

Of the 16,270 main-set entries with a reliable DESI classification, 99.14% are extragalactic and 140 are stars. That figure overstates the purity of the catalogue. Every matched row is a deliberate DESI target, and DESI chose its targets from this same Legacy photometry with cuts (BGS, LRG, ELG) that are themselves galaxy selections. The match therefore measures how often DESI's own galaxy selection picks galaxies, a number its cuts largely decide. Those 16,270 objects are also the ones removed from the catalogue as already identified, so the figure describes entries that have left it. We measure the purity of the 161,721 that remain separately, by visual inspection (Section 5.6).

Of our entries, 63.5% appear in VII/292 (Duncan 2022), the published Legacy Surveys photometric-redshift catalogue from which we draw 102,274 of our redshifts, and it assigns $p_{\rm star} < 0.5$ to 85.4% of them. A published catalogue has therefore already treated most of these sources as non-stellar. We do not count that as identification either, because VII/292 assigns a photometric redshift and a star probability to every source in the parent imaging, stars and artefacts included, and NED and SIMBAD ingest none of its entries as extragalactic classifications. DEBBIE likewise classifies sources that Tractor had already detected and adds no detections of its own.

VST ATLAS DR4 (Shanks et al. 2025) classes 98.86% of the DEBBIE entries it detects as galaxies (Section 5.6), but it was not searched. The screen counts automated galaxy classes only from the catalogues it searched,

such as the SDSS DR12 photometric types and GLADE+, so the novelty claim holds for the catalogues and services above, and a wider search would find an automated class for most entries.

The screening depends most on NED, which aggregates the identifications of most published extragalactic surveys, and we searched no catalogue outside these services. SMUDGes (Zaritsky et al. 2019) has already designated about 7,000 candidates found in this same imaging.

5.4 Physical properties

Redshifts are available for 99.74% of the catalogue, and all are photometric, because the only spectroscopic redshifts in the candidate list were DESI's and they left with the DESI-identified entries. Every distance-dependent quantity below therefore depends on photometric redshifts. Of these, 102,274 come from the Legacy Surveys DR8 photometric-redshift catalogue (VizieR VII/292, Duncan 2022), matched through the CDS X-Match service (Boch et al. 2012), and 59,020 from a gradient-boosted photometric-redshift model, fitted with scikit-learn (Pedregosa et al. 2011), that we trained on our own $grzW1$ photometry for sky outside the public photo-z coverage. Against the Legacy photo-z on spatially held-out sky this estimator has $\sigma_{\mathrm{NMAD}} = 0.046$, and a shuffled-label control degrades to 0.194. We record the photo-z source per object and keep the two estimates apart in any analysis that depends on redshift quality.

Distances assume flat ΛCDM with $H_0 = 70$ km s^{-1} Mpc^{-1} and $\Omega_m = 0.3$, and absolute magnitudes are not k-corrected. Stellar masses apply the $g{-}r$ mass-to-light relation of Bell et al. (2003) to observed colours, which is sound only at low redshift. Below $z = 0.2$ they are factor-of-two estimates. At $z \geq 0.35$ the observed colour is far redder than the rest-frame one, so the masses come out too high, 42% of them above $10^{12} M_\odot$, and they should not be used there.

The median catalogue object lies at $z = 0.294$ with $r_{\mathrm{eff}} = 16$ kpc and $M_g = -19.95$ (Fig. 4), which are the parameters of an ordinary galaxy. Cosmological surface-brightness dimming explains why such galaxies pass a low-surface-brightness cut. The bolometric Tolman factor, $10 \log_{10}(1+z)$, is 1.12 mag arcsec^{-2} at the median redshift. For surface brightness per unit frequency, which is what AB magnitudes measure, the factor before any k-correction is $7.5 \log_{10}(1+z)$, or 0.84 mag arcsec^{-2}. Either is a large part of the 2.5-mag selection window, so a normal galaxy at the median redshift enters the window without being diffuse. The argument holds for the population and should not be applied to a single object.

DEBBIE therefore lists previously uncatalogued galaxy candidates, most of them distant and dimmed. The exception is a nearby tail of 2,591 objects at $z < 0.05$, with median $r_{\mathrm{eff}} = 3.4$ kpc and median $\log(M_\star/M_\odot) = 8.9$, the sizes and masses of dwarfs. Of these, 2,305 satisfy $r_{\mathrm{eff}} > 1.5$ kpc and mean $\mu_{\mathrm{eff}} > 24$ mag arcsec^{-2}, which approximates the ultra-diffuse criteria of Dokkum et al. (2015); those criteria are conventionally stated in central surface brightness, and ours is the mean.

5.5 Selection function

We measure the selection function on real catalogued galaxies in place of synthetic injections. For each known galaxy inside the observed footprint we ask whether the sweeps contain a resolved counterpart and whether the photometric cuts retain it, so real profiles, neighbours and imaging defects enter at their true rates. The measurement covers the cuts only. It excludes the novelty screen and the fragment and clump filters, whose losses Section 5.5.1 measures.

The reference population is the non-stellar entries of the 110-million-object screening index, restricted to $|b| \geq 30°$ and to sky the extract covers, and matched at 3″. Of 235,715 such galaxies, 33,595 (14.2%) appear in the source extract as resolved sources with $r_{\mathrm{eff}} \geq 2''$. Most of those lie outside the surface-brightness window this catalogue targets, and across all of them the cuts retain 10.4%. In 15 of the 17 cells of the size and surface-brightness grid inside the selection window that hold at least 30 reference galaxies, the photometric cuts alone retain between 70% and 89% of them (Fig. 5b). The other two cells retain 56% (8–10″, μ 24–25) and 63% (10–15″, μ 24–25). Wilson 95% intervals are 0.1–9 percentage points (half-width) for cells with at least 50 galaxies and are given cell by cell in `selection_function.json` in the release.

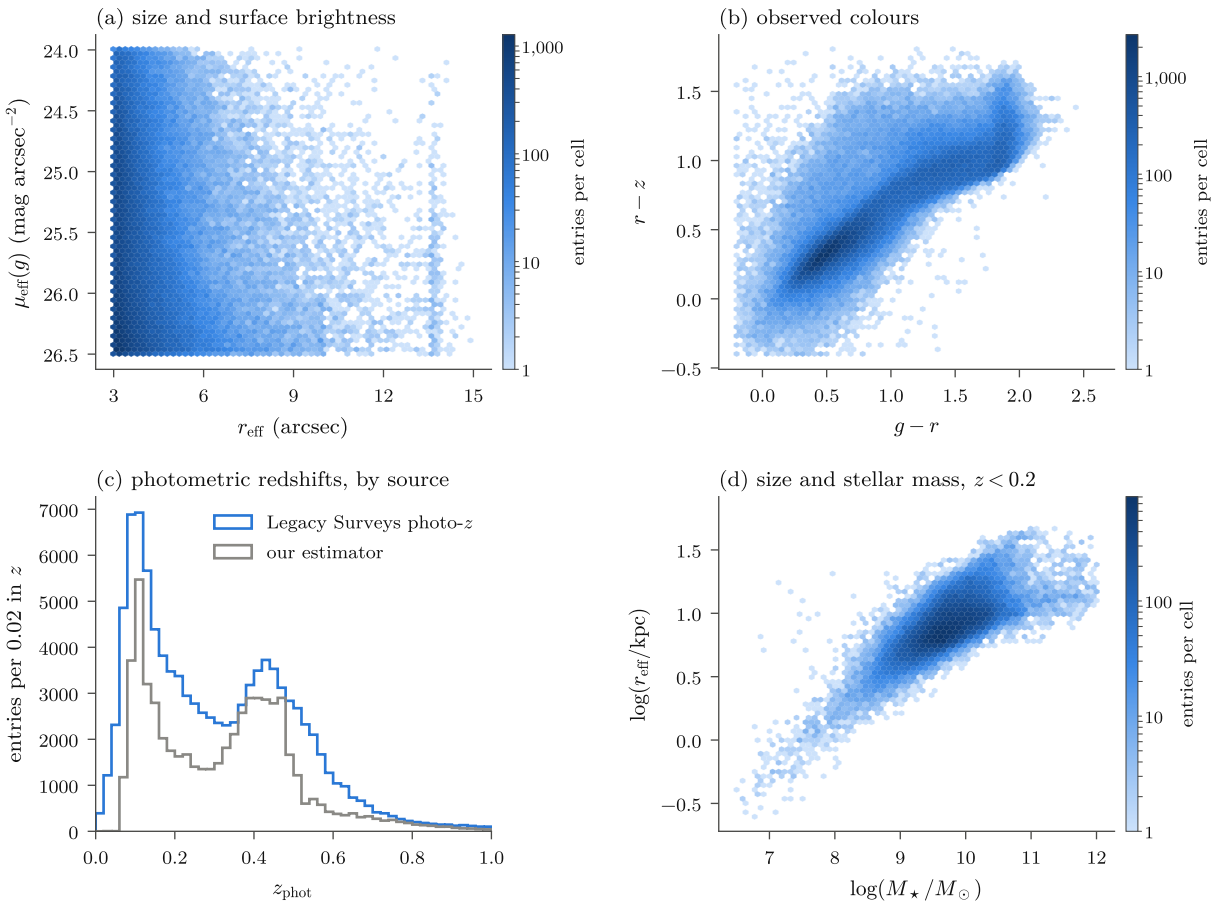


Figure 4: What the catalogue contains: mostly ordinary galaxies dimmed by distance, with median photometric redshift 0.29. The released quantities for the 161,721 entries: (a) effective radius and mean g surface brightness, where the vertical stripe is 148 entries with $r_{\rm eff}$ of exactly 13.624″ (52 pixels); (b) observed colours; (c) photometric redshifts by source; (d) physical size and stellar mass for entries at $z < 0.2$, where the mass estimate holds.

Above $r_{\rm eff} = 15''$ one of 34 known galaxies survives. With $N = 34$ the Wilson 95% interval runs from 0.5% to 14.9% around a point estimate of 2.9%, so this bounds the retention without measuring it. The cause is that Tractor splits the largest galaxies into fragments, none of which carries the size and surface brightness of the whole galaxy, so the cuts reject them. DEBBIE is substantially incomplete above $r_{\rm eff} \approx 15''$ and cannot constrain the abundance of large LSB systems.

5.5.1 Losses in the screen

The selection function above is not end to end. The screen removed most candidates while the sweeps were mined, before any candidate list existed, and recorded nothing about them, so we counted every step again by applying the cuts to the source extract (Fig. 1 gives each count). Of the 1,254,535 sources that pass the cuts at $|b| \geq 30^\circ$, 66% (824,113) lie within $20''$ of an object in the local index, and in 289,138 of those matches the object is an SDSS galaxy within $5''$: SDSS had already catalogued the candidate itself. The fragment filter removes 122,489 and the clump filter 16,405. Rules used early in the mining removed 3,966 more: 1,939 by a fragment radius of $90''$, since reduced to $45''$, and 2,027 that left no record. The queries never reached 5,584. That leaves 281,978 candidates queried, of which NED, SIMBAD or HyperLEDA identify 100,369, and merging at export leaves 177,991 before the DESI step.

The fragment filter is the loss that matters for completeness: besides the fragments it is meant for, it removes a genuine candidate whenever an unrelated catalogued object lies 20–$45''$ away. Among candidates that no catalogue lists within $20''$ it removes 29%, almost independent of size: 29% at $r_{\rm eff} = 3$–$4''$ and 26% at 10–$15''$. End to end, the selection fraction is roughly the cut retention times the share the filter keeps: near 10–$15''$ and $\mu_{\rm eff}$ 24–25, $0.63 \times 0.73 \approx 0.46$. The incompleteness above $r_{\rm eff} \approx 15''$ comes from Tractor's fragmentation of large galaxies and says little about the sky. We release the 12,083 candidates with $r_{\rm eff} \geq 6''$ that the local screen removed from the candidate list, with the extract, so the removals can be checked.

The selection function is completeness relative to *known* galaxies, so a population absent from both the published catalogues and the sweeps cannot appear in it. It also cannot be split into detection and selection terms, because the archived source extract keeps only sources with $r_{\rm eff} \geq 2''$, and we did not keep the raw sweeps that would allow the split.

The catalogue is also non-uniform on the sky (Fig. 5a), and nearly empty inside the SDSS imaging footprint. Between right ascensions 150° and 200° it holds 14.8 entries per deg^2 at $-10^\circ < \delta < -4^\circ$, south of the footprint, and 0.03 at $+4^\circ < \delta < +20^\circ$, inside it. SDSS DR12 lists about 17,000 photometric galaxies per deg^2 to $r = 22.5$ there, so a random position has on average 1.6 within the $20''$ screening radius and 6.5 in the 20–$45''$ fragment annulus. Most candidates there were catalogued before: 80% of those inside the footprint have an SDSS galaxy within $5''$, the candidate itself, and the fragment filter removes nearly all of the rest. DEBBIE therefore says almost nothing about that sky. Density also falls by a factor of two to four in the band $-2^\circ < \delta < +2^\circ$ (1,405 and 1,571 objects in the two equatorial 2° bins, against 3,006–5,444 in the six neighbouring bins), where prior cataloguing is densest (the SDSS equatorial stripes, including Stripe 82). Clustering analyses must model both.

5.6 Purity

Visual inspection of a stratified sample finds a galaxy at the catalogued position for 89.1% of entries (95% CI 86.4–91.6%). That is 93.7% (88.9–97.5%) for the 63,272 entries with $\mu_{\rm eff} < 25.5$ mag arcsec^{-2} and 86.0% (83.0–89.0%) for the 98,449 fainter ones.

We drew a sample of 1,999 objects stratified over 28 cells of the ($r_{\rm eff}$, $\mu_{\rm eff}$) plane, with an effective size of 1,776, the number of equally weighted objects that would give the same precision. Each object carries the number of catalogue entries it stands for, so every figure here is a weighted reconstruction of the catalogue. The sample was put in random order after it was drawn, and we inspected its first 999 objects, a random subsample covering every cell, in two stages. In the first, on $75''$ colour cutouts at $0.375''$ per pixel (the scale of Fig. 6), we accepted 601 objects as galaxies and set 398 aside for closer inspection. Each object set aside

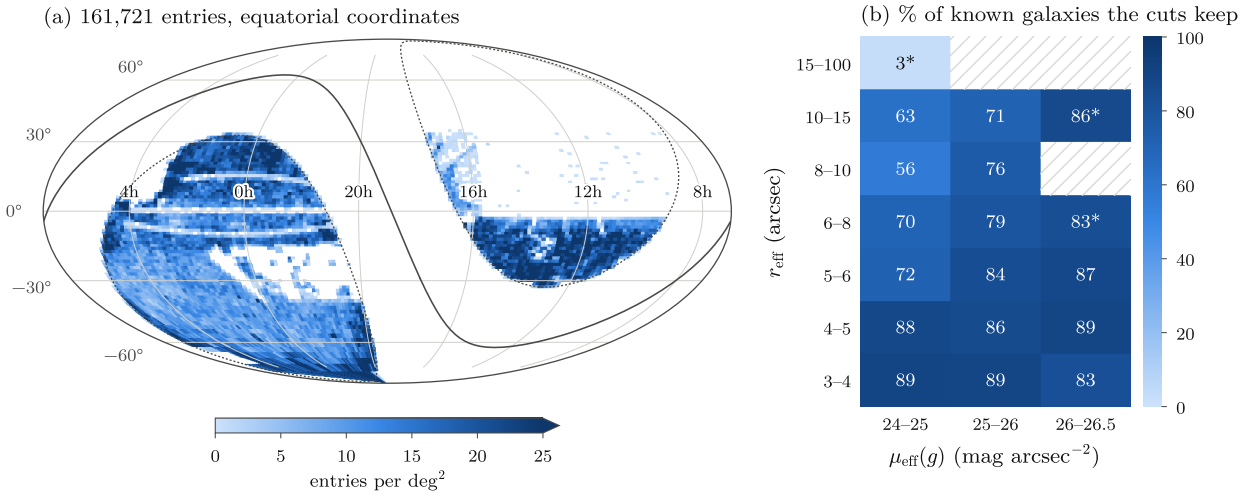


Figure 5: Where the catalogue is, and how much of the known galaxy population its cuts keep. (a) Entries per square degree in equal-area cells, right ascension increasing to the left; solid line, the Galactic plane; dotted lines, $|b| = 30°$. North of the equator between 8h and 16h the catalogue is nearly empty, because SDSS had already catalogued most candidates there. (b) The percentage of known galaxies present in the sweeps that the photometric cuts keep, by size and surface brightness: 56–89% from 3 to 15″, and 1 of 34 above 15″. Hatched cells hold fewer than 30 reference galaxies, starred cells fewer than 50.

was then examined in a 30″ view at 0.15″ per pixel beside the Tractor model and the residual, and judged galaxy, non-galaxy or undecided.

The closer inspection found a galaxy in 264 of the 398 objects set aside and none in 83, and left 51 undecided. Re-examining only the objects set aside could only raise the purity, because an accepted object that is not a galaxy would never be caught, so 100 accepted objects went through the closer inspection as well, in the same sitting, interleaved with the rest and unlabelled. Of the 99 that got a decision, 2 were rejected. We apply that rate, within each surface-brightness half, to the accepted objects that were not inspected again. Intervals come from a bootstrap over the sample objects, with the rate estimated afresh in each resample.

Altogether 52 objects remain undecided, the 51 above and one of the re-examined accepted objects, and are left out of these percentages; counting every one of them as a non-galaxy gives 83.7%, and as a galaxy 89.8%. A single inspector made every verdict, so the intervals carry no term for disagreement between inspectors. The inspection recorded only whether a galaxy was present, so the rejected objects are an unsorted mix of stars, artefacts, noise and fragments of larger sources.

VST ATLAS DR4 images part of this footprint with a different telescope, pipeline and seeing, to a point-source limit of $r = 22.7$, and catalogues everything it detects, stars and noise included. Matching 20,000 random catalogue positions at 2″ finds a counterpart for 28.5%, and of those 98.86% carry `MERGEDCLASS` galaxy or probable galaxy, 0.98% star or probable star, and 0.16% noise. That 98.86% is far above the inspected purity, and the inspected sample shows why. Inside the VST footprint the inspection found 29 objects that are not galaxies; VST detects 21 of them and classes 20 of those as galaxies. At these magnitudes VST calls nearly everything it detects a galaxy.

5.6.1 Stars

Stars can be identified from the catalogue's own photometry, and the released tables flag them and keep them. We trained a gradient-boosted classifier (XGBoost) on 21 Tractor quantities: the effective radius, its inverse variance and its signal-to-noise; the signal-to-noise of the g and r fluxes; the difference and ratio in χ^2 between the Sérsic and point-source fits; the Sérsic index; the two ellipticity components and their modulus; the fraction of flux from neighbours in g and r; the fractions inside masks and inside the fit in g; Galactic

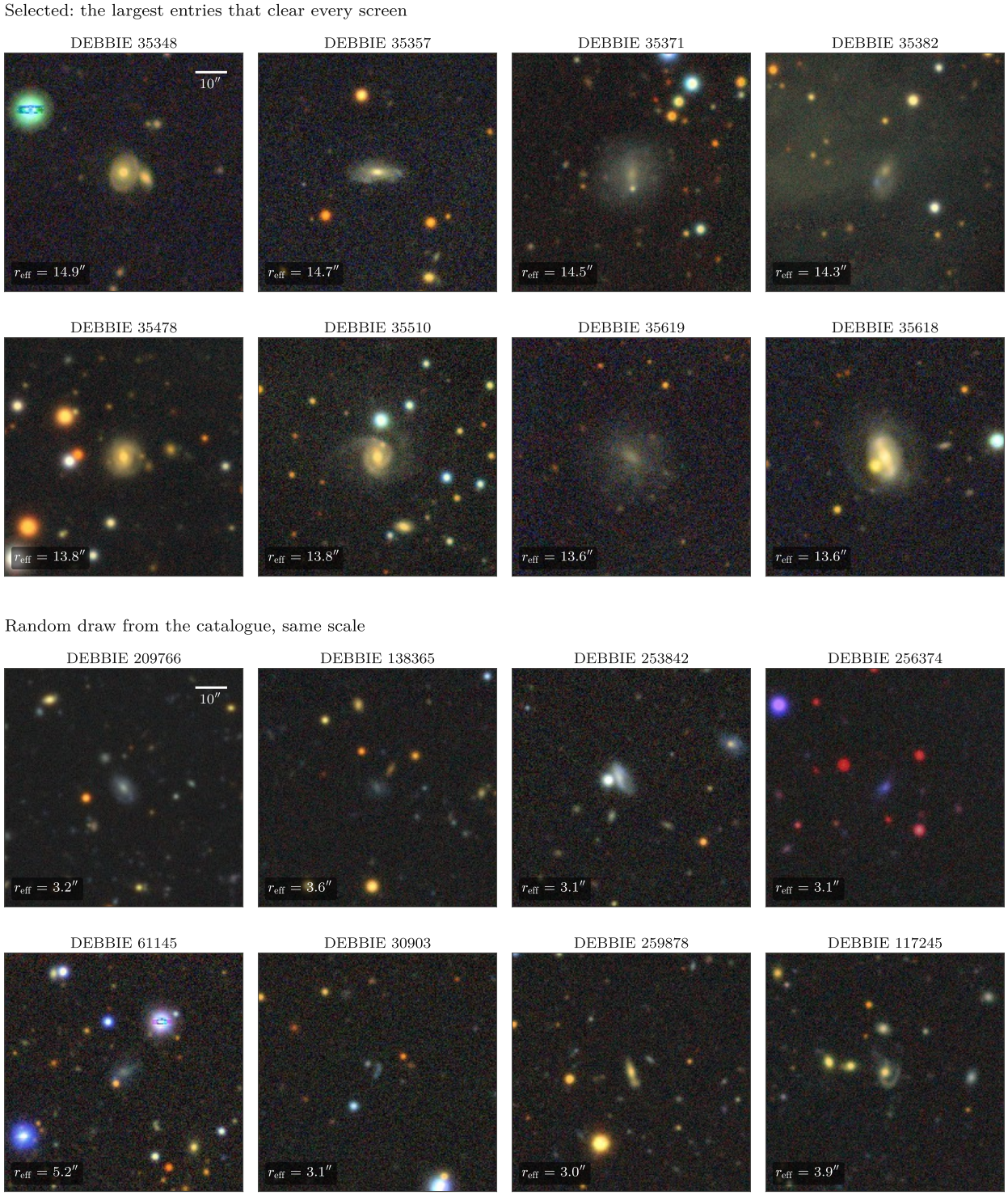


Figure 6: What the entries look like, at the scale of the first inspection: 75″ per panel at 0.375″ per pixel, each labelled with its designation and effective radius. Upper block: the eight largest entries that clear every screen. Lower block: eight drawn at random. The random draw is smaller, fainter and more marginal, and it is what the catalogue typically contains. Saturated stars and fields with a missing band appear as the survey images show them. Images: Legacy Surveys / D. Lang (Perimeter Institute).

reddening; the numbers of g and r exposures; and the $g-r$, $r-z$ and $r-W1$ colours. The training set is the entries DESI DR1 observed, 22,972 with a galaxy spectrum at $z \geq 0.005$ against 338 with a stellar one. All of them passed the catalogue's selection, and none is in the catalogue, because the DESI-identified entries were removed. Out of fold the classifier reaches AUC 0.990, against 0.790 for the Legacy Surveys star probability $p_{\rm star}$ (Duncan 2022). At the threshold that flags 1.0% of the spectroscopic galaxies it flags 86.1% of the spectroscopic stars.

DESI chose which stars to observe, so we tested the classifier on stars it had not been trained on. Gaia DR3 (Gaia Collaboration et al. 2016, 2023), which we use nowhere else in this work, has a source within 2″ of 1.69% of 40,000 random catalogue positions, and 0.74% (95% CI 0.66–0.83%) show a 5σ parallax or proper motion and so are stars. That is a lower bound on the stellar fraction, since Gaia astrometry reaches only the brighter entries. At the same threshold the classifier flags 85.8% of those 296 entries, against 86.1% of the DESI stars it was trained on.

For these sources $p_{\rm star}$ is not a calibrated probability. Its mean over the spectroscopic galaxies is 0.090, against 0.178 over the 63.5% of catalogue entries that have one. Summing it over the catalogue to count stars therefore overstates them, because galaxies alone would supply 50% of that sum. A value above 0.9 is still informative, since only 1.5% of the spectroscopic galaxies reach it, and we use it as a second, weaker flag for entries the classifier passes. Together the two flag 89.0% of the DESI stars, 92.2% of the Gaia stars and 2.5% of the spectroscopic galaxies.

Each entry carries the classifier's star score and a flag: 1 for the 3,708 entries above the classifier threshold, and 2 for the 4,847 further entries with $p_{\rm star} > 0.9$. Together they mark 5.3% of the catalogue. Large galaxies are flagged more often than small ones, 7.0% of the spectroscopic galaxies with $r_{\rm eff} \geq 6''$ against 1.7% of those with $r_{\rm eff} < 4''$. Discarding flagged entries would therefore remove large galaxies preferentially, so the selection function of Section 5.5 describes the catalogue with them kept.

Stars are a small part of the contamination, and on the inspected sample the flag does not separate galaxies from the rest. It marks 5 of the 85 objects the closer inspection rejected and 39 of the 862 galaxies. Leaving flagged entries out moves the purity by 0.01 percentage points.

5.6.2 Two-model flag

The classifiers of Section 4 were trained on profiles measured before the Legacy Surveys cutout service changed its handling of small cutouts, and the catalogue was measured after. We therefore re-extracted all 21,052 labelled objects behind the benchmark, of every class, through the path the catalogue uses. The single-split AUC is 0.9813, against 0.9810 before, and the five-fold AUC of the trees on the full representation is 0.9793, against 0.9790.

On the re-extracted benchmark we fitted the two models that flag catalogue entries: gradient-boosted trees on the radial representation and the bidirectional recurrent network. Neither uses the six catalogue features, which catalogue entries do not have. Over five folds the trees score 0.9795, the network 0.9768 and the mean of their probabilities 0.9808. Each catalogue entry is scored with that mean.

Against the 862 galaxies and 85 non-galaxies of the inspected sample, the mean separates the two with AUC 0.846 (0.810–0.879); the trees alone reach 0.822 and the network 0.837. It also finds what the star flag misses: against only the 80 non-galaxies the star flag passes, the AUC is 0.841. The same pair trained on a different set (catalogued galaxies against stars, and against Tractor fits fainter than the selection window) scores 0.41, below chance, so the training set matters more here than the architecture.

We flag entries that score below the weighted 5th percentile of the inspected galaxies' scores, so that the flag keeps 95% of galaxies (Fig. 7). The threshold is cross-fitted: we split the sample at random and set each half's threshold on the other half, so no object is flagged by a threshold its own verdict helped set. By the sample weights the flag marks 7.2% of entries, 5.0% of galaxies and 22% of non-galaxies. Among the entries it passes the purity is 90.7% (88.0–93.2%), 1.6 percentage points (0.7–2.6) above the whole catalogue measured on the same resamples. The whole improvement lies in the faint half, which rises from 86.0% to

88.5%, while the bright half stays at 93.7%. A flagged entry is a galaxy 67% of the time (54–79%). The flag removes nothing, so the selection function and the purity above describe the catalogue as released.

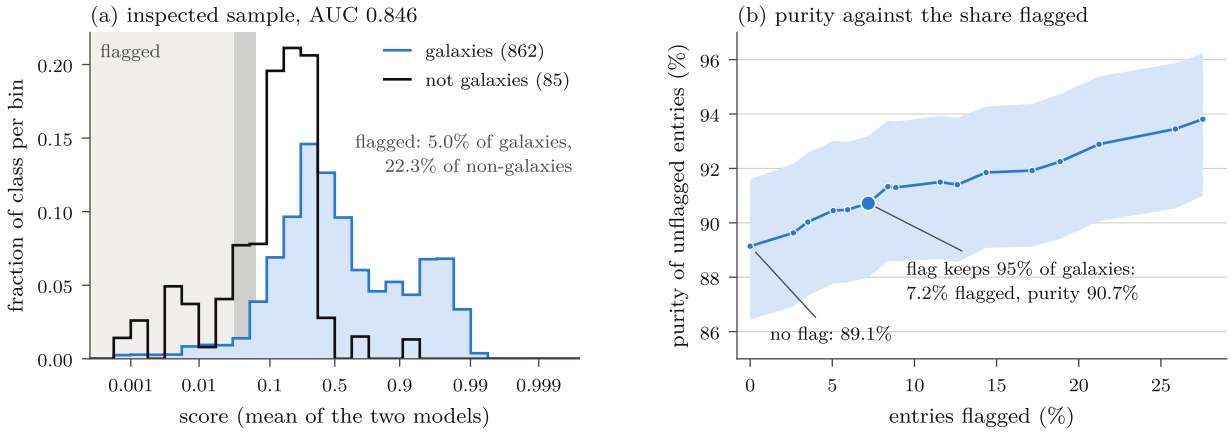


Figure 7: The two-model flag on the inspected sample. (a) Scores of the 862 galaxies and 85 non-galaxies, each class weighted to the catalogue; the scores overlap, but the non-galaxies sit lower (AUC 0.846). Entries below the threshold are flagged; the darker band spans the two cross-fitted thresholds. (b) Purity among the entries the flag passes, against the share of entries it flags, with 95% bootstrap intervals. Each point keeps a different fraction of the inspected galaxies, from all of them to 80%; the large point is the flag used here, which keeps 95%.

6 Comparison with DES and SMUDGes

The two published catalogues most similar to DEBBIE reached their purity in two steps: an automated pipeline, then a visual check of every candidate it passed. No DEBBIE entry was inspected individually, so the published figures for their automated step are the ones to set beside DEBBIE's (Table 2).

Table 2: Without a visual check of each entry, DEBBIE's purity is close to the SMUDGes pipeline's before its visual check and well above the DES pipeline's, at 7 and 23 times their size. Surface brightnesses in mag arcsec^{-2}; $r_{1/2}$ and r_e are half-light radii and $\mu_{0,g}$ is the central g-band surface brightness, each as its paper defines it. DES LSBG: Tanoglidis, Drlica-Wagner, et al. (2021), who estimate that 44% of the objects their classifier passed are false positives, and their visual inspection agreed. SMUDGes V: Zaritsky et al. (2023); 105 of 1,374 candidates were judged false positives at visual review, and Papers II and III, quoted there, found 2.6% and 2.8%, so 97.4% and 97.2% purity in those releases. DEBBIE: inspected stratified sample, with and without the two-model flag (Section 5.6.2).

catalogue	imaging	selection	purity before the visual check	entries
DES LSBG	DES Y3	$r_{1/2} > 2.5''$, $\bar{\mu}_{\rm eff}(g) > 24.2$	about 56%	23,790
SMUDGes V	Legacy Surveys DR9	$\mu_{0,g} \geq 24$, $r_e \geq 5.3''$	92.4%	7,070
DEBBIE	Legacy Surveys DR10	$r_{\rm eff} \geq 3''$, $24 \leq \mu_{\rm eff}(g) \leq 26.5$	89.1%, or 90.7% unflagged	161,721

The populations behind these numbers differ: DES keeps low-surface-brightness galaxies and SMUDGes large diffuse ones, while most DEBBIE entries are ordinary galaxies at $z \approx 0.3$ dimmed by distance (Section 5). Purity is also counted differently: theirs from candidates an inspector rejected, ours from sampled entries with a galaxy at the catalogued position. Both catalogues then removed what their inspectors rejected, so

their final purity should exceed DEBBIE's. What DEBBIE shows is that an automated selection can reach the purity those pipelines had before that step, at 7 and 23 times their size, with the purity measured on an inspected sample.

7 Summary

Light profiles in concentric annuli, classified with gradient-boosted trees, match a published convolutional network on its own benchmark. With a recurrent network they flag doubtful entries in DEBBIE, a catalogue too large to inspect entry by entry, whose purity is measured on an inspected sample.

1. Gradient-boosted trees on annular light profiles reach AUC 0.9810 on the DeepShadows benchmark labels, against 0.9098 for six conventional catalogue features with the same classifier. Over five folds the paired improvement is $+0.067 \pm 0.008$, and averaging the trees with a recurrent network adds +0.0017.
2. DEBBIE holds 161,721 galaxy candidates over about 12,900 deg^2, nearly none inside the SDSS imaging footprint, with no prior identification as galaxies in HyperLEDA, SDSS DR12, GLADE+, the 2MASS XSC, 2dFGRS, 6dFGS, SMUDGes, NED, SIMBAD or DESI DR1.
3. A galaxy lies at the catalogued position for 89.1% of entries (86.4–91.6%), from visual inspection of a stratified sample in two stages.
4. Boosted trees and a bidirectional recurrent network, retrained on the re-extracted benchmark, separate the inspected galaxies from non-galaxies with AUC 0.846. Flagging the lowest-scoring 7.2% of entries raises purity among the rest to 90.7%.
5. Without inspecting each entry, that purity is close to the SMUDGes V pipeline's before its visual check (92.4%) and well above the DES search's at the same stage (about 56%), at 7 and 23 times the size of those catalogues.
6. A star flag trained on DESI spectra marks 85.8% of Gaia stars it was not trained on, at the threshold that marks 1.0% of spectroscopic galaxies.
7. Inside the selection window the photometric cuts retain 56–89% of known resolved galaxies, and the fragment filter removes 29% of the candidates no catalogue lists.
8. The catalogue is substantially incomplete above $r_{\rm eff} \approx 15''$.
9. All redshifts are photometric. The median entry lies at $z = 0.29$, so most entries are ordinary galaxies dimmed by distance, and 2,591 entries at $z < 0.05$ have the sizes and masses of dwarfs.

8 Data availability

We release the catalogue as `DEBBIE.csv` (161,721 rows) with `DEBBIE_lowb_extension.csv` (78,713 rows), version 1.4, each with an ECSV copy and a SHA-256 checksum. The tables, the code and the source table will be deposited at Zenodo, and the journal version carries both tables in machine-readable form. Table 3 shows the first entries and Table 4 lists the columns. Designations take the form DEBBIE NNNNNN, a running number of up to six digits written without leading zeros (DEBBIE 3 to DEBBIE 268380), and each entry also carries a positional name of the form DEBBIE Jhhmmss.s±ddmmss. A number is assigned once and never reused, so designations are stable across versions, and the sequence has gaps where merged duplicates retired their numbers. The full selection can be recomputed from the released 3.2 GB distilled source table (33.5 million resolved sources, 31 columns), which reproduces the catalogue exactly.

Table 3: The first five entries of DEBBIE, by designation number. Positions in degrees, $r_{\rm eff}$ in arcsec, $\mu_{\rm eff}(g)$ in mag arcsec^{-2}, g and $g-r$ in mag. The full table, with positional names and every column of Table 4, is available in machine-readable form.

designation	RA	Dec	$r_{\rm eff}$	$\mu_{\rm eff}(g)$	g	$g-r$	$z_{\rm phot}$	star flag
DEBBIE 3	185.45112	−8.85253	13.83	25.20	17.50	0.88	0.071	0
DEBBIE 5	12.92114	−6.03002	13.83	25.52	17.82	0.39	0.026	0
DEBBIE 6	9.32609	29.07681	13.83	24.90	17.20	0.53	0.034	0
DEBBIE 9	157.94469	−8.17343	13.62	25.63	17.96	0.56	0.045	0
DEBBIE 10	168.10858	−8.55186	13.62	25.38	17.71	0.53	0.016	0

Table 4: Columns of the released tables.

column	unit	contents
`name`, `iau`		designation (DEBBIE N) and coordinate-based name
`ra_deg`, `dec_deg`	deg	position from the DR10 sweeps
`position`		the same position in sexagesimal form
`r_eff_arcsec`	arcsec	Tractor effective radius
`mu_eff_g`	mag arcsec^{-2}	mean g surface brightness within $r_{\rm eff}$
`g_mag`, `g_r`, `r_z`	mag	g magnitude and colours from Tractor fluxes, not corrected for Galactic extinction
`sersic`		Tractor Sérsic index
`z_phot`, `z_source`		photometric redshift and its source (`photo` for VII/292, `own` for ours)
`r_eff_kpc`, `M_g`, `log_mstar`	kpc, mag, dex	physical size, absolute magnitude and stellar mass, where a redshift exists; the mass is usable only at $z < 0.2$
`pstar`		Legacy Surveys star probability from VII/292
`star_score`, `star_flag`		star score and flag of Section 5.6.1
`prior_designations`		designations from automated surveys, from NED

9 Limitations

The photometry is Tractor's (Lang et al. 2016), taken as published. Every redshift is photometric, and 37% come from our own estimator, which is trained on photometric labels. That estimator's quoted scatter is measured against the Legacy photo-z and has not been tested against spectroscopy, so those values should be used only for population statistics. Purity depends on one inspector's verdicts for 999 objects (Section 5.6). One radius, $r_{\rm eff} = 13.624''$, is recorded for 148 entries, which suggests a cap on the radius in the Tractor fit. The selection inherits Tractor's detection incompleteness, fails above $r_{\rm eff} \approx 15''$, excludes $|b| < 30^\circ$ and is non-uniform on the sky where prior surveys are deepest. Inside the SDSS imaging footprint it is nearly empty, because SDSS had already catalogued most candidates there (Section 5.5).

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Facilities: Blanco (DECam), WISE, Gaia, Astro Data Lab.

Software: Astropy (Astropy Collaboration et al. 2022), Matplotlib (Hunter 2007), NumPy (Harris et al. 2020), scikit-learn (Pedregosa et al. 2011), SciPy (Virtanen et al. 2020), the Tractor (Lang et al. 2016), XGBoost (Chen and Guestrin 2016).

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